Single_Cell_Workflow ES data

Load Main Libraries for the Analysis

```
## Loading required package: ggplot2

##
## Attaching package: 'plotly'

## The following object is masked from 'package:ggplot2':
##
##     last_plot

## The following object is masked from 'package:stats':
##
##     filter

## The following object is masked from 'package:graphics':
##
##     layout

## Loading required package: limma

## Loading required package: AnnotationDbi

## Loading required package: stats4

## Loading required package: BiocGenerics

##
## Attaching package: 'BiocGenerics'

## The following object is masked from 'package:limma':
##
##     plotMA

## The following objects are masked from 'package:stats':
##
##     IQR, mad, sd, var, xtabs

## The following objects are masked from 'package:base':
##
##     anyDuplicated, aperm, append, as.data.frame, basename, cbind,
##     colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
##     get, grep, grepl, intersect, is.unsorted, lapply, Map, mapply,
##     match, mget, order, paste, pmax, pmax.int, pmin, pmin.int,
##     Position, rank, rbind, Reduce, rownames, sapply, setdiff, sort,
##     table, tapply, union, unique, unsplit, which.max, which.min

## Loading required package: Biobase

## Welcome to Bioconductor
##
##     Vignettes contain introductory material; view with
##     'browseVignettes()'. To cite Bioconductor, see
##     'citation("Biobase")', and for packages 'citation("pkgname")'.
```

```
## Loading required package: IRanges

## Loading required package: S4Vectors

##
## Attaching package: 'S4Vectors'

## The following object is masked from 'package:plotly':
##
##     rename

## The following objects are masked from 'package:base':
##
##     expand.grid, I, unname

##
## Attaching package: 'IRanges'

## The following object is masked from 'package:plotly':
##
##     slice

##
## Attaching package: 'AnnotationDbi'

## The following object is masked from 'package:plotly':
##
##     select

##

##
## Attaching package: 'gplots'

## The following object is masked from 'package:IRanges':
##
##     space

## The following object is masked from 'package:S4Vectors':
##
##     space

## The following object is masked from 'package:stats':
##
##     lowess

## Loading required package: flexmix

## Loading required package: lattice

##
## Attaching package: 'gridExtra'

## The following object is masked from 'package:Biobase':
##
##     combine

## The following object is masked from 'package:BiocGenerics':
##
##     combine

## Loading required package: SummarizedExperiment

## Loading required package: MatrixGenerics
```

```
## Loading required package: matrixStats

##
## Attaching package: 'matrixStats'

## The following objects are masked from 'package:Biobase':
##
##     anyMissing, rowMedians

##
## Attaching package: 'MatrixGenerics'

## The following objects are masked from 'package:matrixStats':
##
##     colAlls, colAnyNAs, colAnys, colAvgsPerRowSet, colCollapse,
##     colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
##     colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
##     colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
##     colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
##     colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
##     colWeightedMeans, colWeightedMedians, colWeightedSds,
##     colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgsPerColSet,
##     rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
##     rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
##     rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
##     rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
##     rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
##     rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
##     rowWeightedSds, rowWeightedVars

## The following object is masked from 'package:Biobase':
##
##     rowMedians

## Loading required package: GenomicRanges

## Loading required package: GenomeInfoDb

##
## Attaching package: 'SingleCellExperiment'

## The following object is masked from 'package:edgeR':
##
##     cpm

## Loading required package: scuttle

##
## Attaching package: 'scater'

## The following object is masked from 'package:limma':
##
##     plotMDS

## Loading required package: XML

## [1] "Just set global variable and functions here..."
## [1] "Set globals"

## [1] "/home/guest101/FGT2023/t9"
```

Read Kallisto Counts and Associated Input Data (Spikes and Mitochondrial Gene IDs)

The dataset contains ERCC spikes and these spikes are identified as beginning with "ERCC".

Mitochondrial genes are loaded using a short list of mitochondrially encoded mouse genes from Ensembl. These spikes are then used to calculate QC metrics from the files together with library size feature count metrics and total library size for each cell. Cells were excluded if feature counts,library size, mitochondrial reads or ERCC spike counts were more than 3 Median Absolute Deviations from the Median of these catagories per cell. Plots were used to check this threshold against the distribution of the data. ***

##Calculate Initial Quality Control Metrics

Select only samples that pass the basic Sample QC criteria * library size must not be less than 3 Median Absolute Deviations from median size * total_features_by_counts not be less than 3 Median Absolute Deviations from median counts * mitochondrial counts must be more than 3 Median Absolute Deviations from median counts * total spike counts must be more than 3 Median Absolute Deviations from median counts

This approach is designed to remove samples which deviate from the median. The risk is that we throw away real biology with this filtering. In practice filtering is checked against biological features in the data to ensure bias is not introduced by this step.

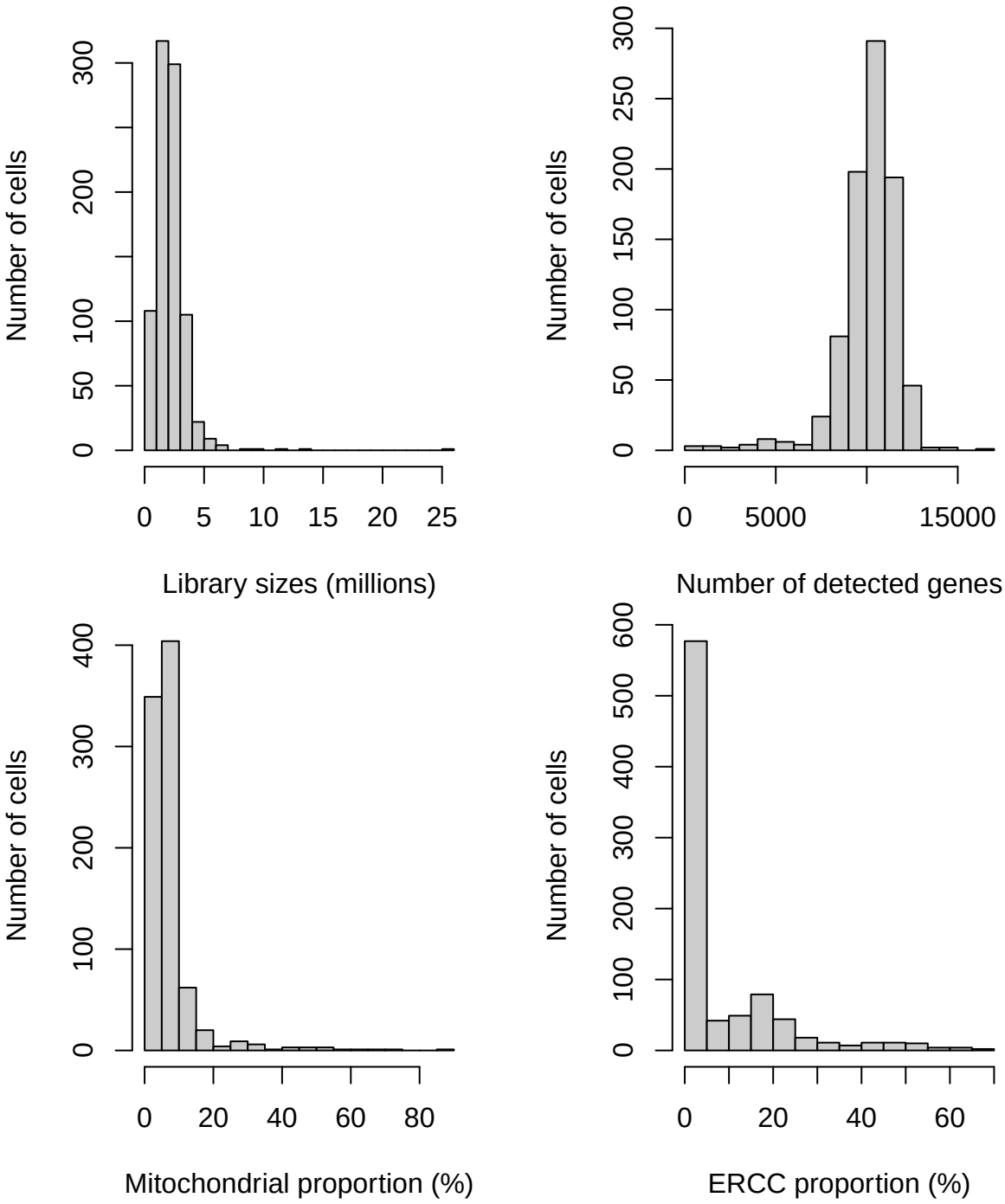
```
## [1] "Checking Internal Controls..."

## [1] "Count Spike: 92"

## [1] "Count Mito: 13"

##  [1] 32632 32633 32634 32635 32636 32637 32638 32639 32640 32641 32642 32643
## [13] 32644 32645 32646 32647 32648 32649 32650 32651 32652 32653 32654 32655
## [25] 32656 32657 32658 32659 32660 32661 32662 32663 32664 32665 32666 32667
## [37] 32668 32669 32670 32671 32672 32673 32674 32675 32676 32677 32678 32679
## [49] 32680 32681 32682 32683 32684 32685 32686 32687 32688 32689 32690 32691
## [61] 32692 32693 32694 32695 32696 32697 32698 32699 32700 32701 32702 32703
## [73] 32704 32705 32706 32707 32708 32709 32710 32711 32712 32713 32714 32715
## [85] 32716 32717 32718 32719 32720 32721 32722 32723

##  [1] 17072 17073 17074 17075 17076 17077 17078 17079 17080 17081 17082 17083
## [13] 17085
```



After Completion of QC Filtering of Individual Cell Libraries

This table summarises the number of elements removed by each filtering step and the number of cell sample libraries remaining in the data. ***

```
data.frame(ByLibSize=sum(libsize.drop), ByFeature=sum(feature.drop),
           ByMito=sum(mito.drop), BySpike=sum(spike.drop), Remaining=ncol(sce))
```

```
##   ByLibSize ByFeature ByMito BySpike Remaining
## 1        39        35     59     307       515
```

```r
print("Dimensions finished object")
```

```
## [1] "Dimensions finished object"
```

```r
dim(sce)
```

```
## [1] 32723   515
```

```r
print("Dimensions finished starting object")
```

```
## [1] "Dimensions finished starting object"
```

```r
dim(sceX)
```

```
## [1] 32723   869
```

```r
colnames(sce)
```

```
##   [1] "ERR1211464" "ERR1211465" "ERR1211467" "ERR1211468" "ERR1211469"
##   [6] "ERR1211470" "ERR1211471" "ERR1211472" "ERR1211473" "ERR1211474"
##  [11] "ERR1211476" "ERR1211480" "ERR1211481" "ERR1211482" "ERR1211483"
##  [16] "ERR1211484" "ERR1211485" "ERR1211486" "ERR1211487" "ERR1211488"
##  [21] "ERR1211489" "ERR1211490" "ERR1211491" "ERR1211492" "ERR1211493"
##  [26] "ERR1211494" "ERR1211495" "ERR1211496" "ERR1211497" "ERR1211498"
##  [31] "ERR1211499" "ERR1211500" "ERR1211501" "ERR1211502" "ERR1211503"
##  [36] "ERR1211505" "ERR1211506" "ERR1211507" "ERR1211508" "ERR1211509"
##  [41] "ERR1211512" "ERR1211513" "ERR1211515" "ERR1211516" "ERR1211517"
##  [46] "ERR1211519" "ERR1211520" "ERR1211521" "ERR1211522" "ERR1211523"
##  [51] "ERR1211524" "ERR1211525" "ERR1211526" "ERR1211527" "ERR1211528"
##  [56] "ERR1211529" "ERR1211530" "ERR1211531" "ERR1211532" "ERR1211533"
##  [61] "ERR1211534" "ERR1211535" "ERR1211536" "ERR1211540" "ERR1211541"
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##  [71] "ERR1211551" "ERR1211552" "ERR1211553" "ERR1211554" "ERR1211557"
##  [76] "ERR1211559" "ERR1211560" "ERR1211561" "ERR1211562" "ERR1211563"
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##  [86] "ERR1211569" "ERR1211570" "ERR1211571" "ERR1211572" "ERR1211573"
##  [91] "ERR1211574" "ERR1211575" "ERR1211576" "ERR1211577" "ERR1211578"
##  [96] "ERR1211579" "ERR1211580" "ERR1211582" "ERR1211583" "ERR1211584"
## [101] "ERR1211585" "ERR1211586" "ERR1211587" "ERR1211588" "ERR1211589"
## [106] "ERR1211590" "ERR1211591" "ERR1211593" "ERR1211594" "ERR1211595"
## [111] "ERR1211597" "ERR1211598" "ERR1211599" "ERR1211600" "ERR1211601"
## [116] "ERR1211603" "ERR1211604" "ERR1211605" "ERR1211606" "ERR1211607"
## [121] "ERR1211608" "ERR1211610" "ERR1211611" "ERR1211612" "ERR1211613"
## [126] "ERR1211614" "ERR1211615" "ERR1211617" "ERR1211618" "ERR1211620"
## [131] "ERR1211621" "ERR1211622" "ERR1211623" "ERR1211624" "ERR1211625"
## [136] "ERR1211626" "ERR1211627" "ERR1211628" "ERR1211629" "ERR1211631"
## [141] "ERR1211632" "ERR1211633" "ERR1211634" "ERR1211635" "ERR1211636"
## [146] "ERR1211637" "ERR1211640" "ERR1211641" "ERR1211642" "ERR1211644"
## [151] "ERR1211645" "ERR1211646" "ERR1211647" "ERR1211649" "ERR1211650"
## [156] "ERR1211651" "ERR1211653" "ERR1211654" "ERR1211655" "ERR522819"
## [161] "ERR522821"  "ERR522823"  "ERR522824"  "ERR522825"  "ERR522826"
## [166] "ERR522827"  "ERR522828"  "ERR522829"  "ERR522830"  "ERR522831"
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## [181] "ERR522842"  "ERR522844"  "ERR522845"  "ERR522846"  "ERR522847"
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## [191] "ERR522853"  "ERR522854"  "ERR522855"  "ERR522856"  "ERR522857"
```

```
## [196] "ERR522858"  "ERR522859"  "ERR522860"  "ERR522861"  "ERR522862"
## [201] "ERR522863"  "ERR522864"  "ERR522865"  "ERR522867"  "ERR522868"
## [206] "ERR522870"  "ERR522871"  "ERR522873"  "ERR522874"  "ERR522875"
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## [221] "ERR522886"  "ERR522887"  "ERR522888"  "ERR522889"  "ERR522890"
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## [241] "ERR522906"  "ERR522907"  "ERR522908"  "ERR522909"  "ERR522910"
## [246] "ERR522911"  "ERR522912"  "ERR522913"  "ERR522915"  "ERR522916"
## [251] "ERR522917"  "ERR522918"  "ERR522919"  "ERR522920"  "ERR522922"
## [256] "ERR522923"  "ERR522924"  "ERR522925"  "ERR522926"  "ERR522927"
## [261] "ERR522928"  "ERR522929"  "ERR522931"  "ERR522932"  "ERR522933"
## [266] "ERR522934"  "ERR522935"  "ERR522936"  "ERR522937"  "ERR522938"
## [271] "ERR522939"  "ERR522940"  "ERR522941"  "ERR522942"  "ERR522943"
## [276] "ERR522944"  "ERR522945"  "ERR522946"  "ERR522947"  "ERR522948"
## [281] "ERR522949"  "ERR522950"  "ERR522951"  "ERR522956"  "ERR522957"
## [286] "ERR522958"  "ERR522959"  "ERR522960"  "ERR522961"  "ERR522962"
## [291] "ERR522963"  "ERR522964"  "ERR522965"  "ERR522966"  "ERR522967"
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## [306] "ERR522978"  "ERR522979"  "ERR522980"  "ERR522981"  "ERR522982"
## [311] "ERR522983"  "ERR522984"  "ERR522985"  "ERR522987"  "ERR522988"
## [316] "ERR522989"  "ERR522990"  "ERR522991"  "ERR522992"  "ERR522993"
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## [326] "ERR523001"  "ERR523002"  "ERR523003"  "ERR523004"  "ERR523005"
## [331] "ERR523006"  "ERR523007"  "ERR523008"  "ERR523009"  "ERR523010"
## [336] "ERR523011"  "ERR523012"  "ERR523013"  "ERR523014"  "ERR523015"
## [341] "ERR523016"  "ERR523017"  "ERR523018"  "ERR523019"  "ERR523020"
## [346] "ERR523021"  "ERR523022"  "ERR523023"  "ERR523024"  "ERR523025"
## [351] "ERR523026"  "ERR523027"  "ERR523029"  "ERR523030"  "ERR523031"
## [356] "ERR523032"  "ERR523033"  "ERR523034"  "ERR523035"  "ERR523036"
## [361] "ERR523037"  "ERR523038"  "ERR523039"  "ERR523040"  "ERR523041"
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## [401] "ERR523080"  "ERR523081"  "ERR523082"  "ERR523084"  "ERR523085"
## [406] "ERR523086"  "ERR523087"  "ERR523088"  "ERR523089"  "ERR523090"
## [411] "ERR523092"  "ERR523093"  "ERR523094"  "ERR523096"  "ERR523098"
## [416] "ERR523099"  "ERR523101"  "ERR523102"  "ERR523103"  "ERR523104"
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## [426] "ERR523111"  "ERR523112"  "ERR523113"  "ERR523114"  "ERR523115"
## [431] "ERR523116"  "ERR523117"  "ERR523118"  "ERR523119"  "ERR523120"
## [436] "ERR523121"  "ERR523122"  "ERR523123"  "ERR523124"  "ERR523125"
## [441] "ERR523126"  "ERR523127"  "ERR523128"  "ERR523129"  "ERR523130"
## [446] "ERR523131"  "ERR523132"  "ERR523133"  "ERR523134"  "ERR523135"
## [451] "ERR523136"  "ERR523138"  "ERR523139"  "ERR523140"  "ERR523141"
## [456] "ERR523142"  "ERR523144"  "ERR523145"  "ERR523146"  "ERR523147"
## [461] "ERR523148"  "ERR523149"  "ERR523150"  "ERR523151"  "ERR523152"
```
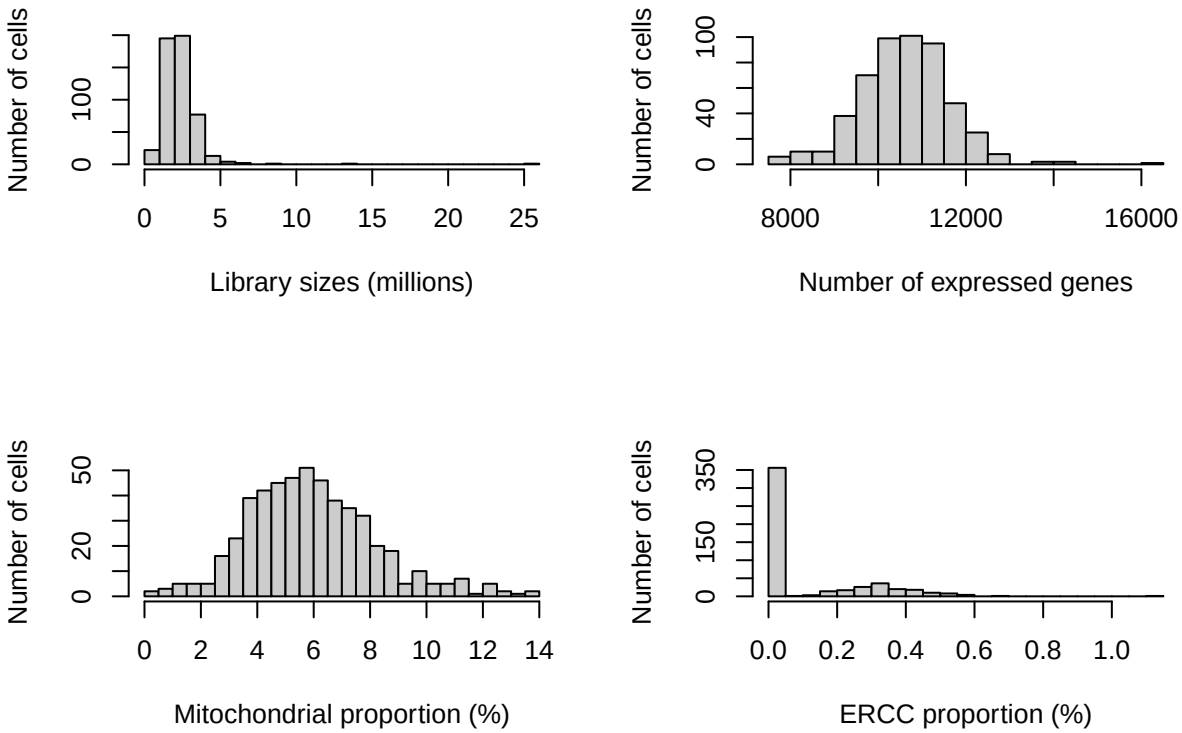

```
## [466] "ERR523153"  "ERR523154"  "ERR523155"  "ERR523156"  "ERR523157"
## [471] "ERR523158"  "ERR523159"  "ERR523160"  "ERR523161"  "ERR523163"
## [476] "ERR523164"  "ERR523165"  "ERR523166"  "ERR523167"  "ERR523168"
## [481] "ERR523169"  "ERR523170"  "ERR523171"  "ERR523172"  "ERR523173"
## [486] "ERR523174"  "ERR523175"  "ERR523178"  "ERR523179"  "ERR523180"
## [491] "ERR523182"  "ERR523183"  "ERR523184"  "ERR523185"  "ERR523186"
## [496] "ERR523187"  "ERR523188"  "ERR523189"  "ERR523190"  "ERR523191"
## [501] "ERR523192"  "ERR523193"  "ERR523194"  "ERR523195"  "ERR523196"
## [506] "ERR523197"  "ERR523198"  "ERR523199"  "ERR523200"  "ERR523201"
## [511] "ERR523202"  "ERR523203"  "ERR523204"  "ERR523205"  "ERR523206"
```

Sample Quality Plots after Removing Poor Quality Samples

This summarises the filtering of samples from the data ***



###

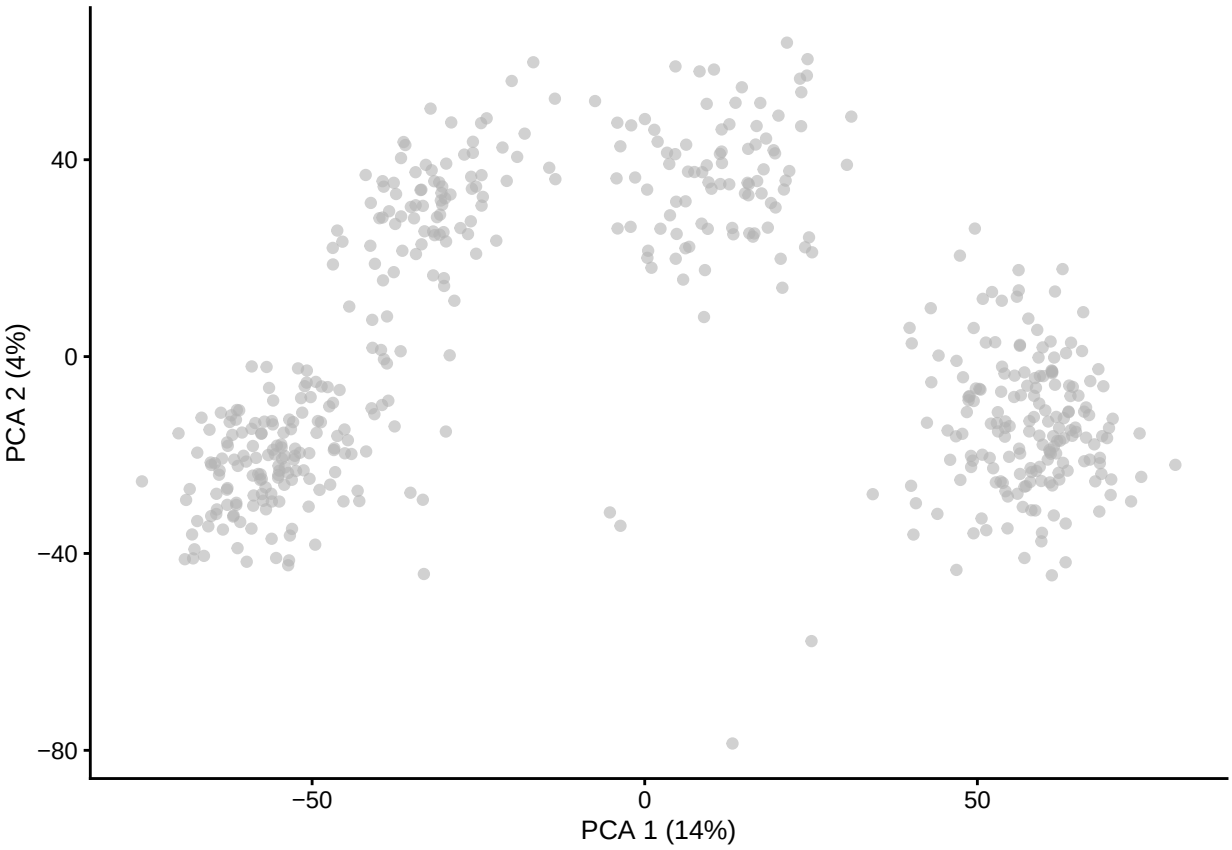
QC filters All Libraries Combined Initial PCA plot using log tranformed input data. Data has not been normalised within R although is normalised to length normalised counts per million during library loading. All genes are included in the plots. Log transformed data has a small count (0.01) added to each count to be normalised to avoid problems with log of 0 values. The PCA plot is calculated off all genes in the samples.

The PCA plot shows two PCA dimensions that account for 18% of the variance. These dimensions seem to separate cell culture conditions (confirmed later). Given this PCA uses all genes in the data set the partitioning of the variance into these components is very promising.

PCA of Raw Log Counts (Just Log Transformed) without Normalisation

This plot just shows the general structure of the prior to normalisation. It uses as input all the data, concluding genes with low or zero counts

```
## [1] "PCA"
```



Conversion to Gene Names from Ensembl IDs as Primary Identifiers

Row names were Ensembl gene names (collapsed from transcripts). Here these identifiers are converted to Gene Symbols using Ensembl annotation. There are one or two genes with ambiguous mappings and so the Ensembl mapping is stored in the ENSEMBL slot and the gene mapping in the SYMBOL slot. This makes the mapping reversible if required. The mapping uses the mouse annotation library available from Bioconductor (org.Mm.eg.db). Row names were switched to gene names (Gene Symbol) for the rest of this analysis.

Note this data has a few genes with ambiguous identifiers- these cannot be further analysed using this approach-primary keys could switch to ENSG_MGI identifiers to account for this but it complicated the analysis for a few 'edge case' genes and so this was not performed here. The genes giving ambigous mapping have been individually checked before exclusion!

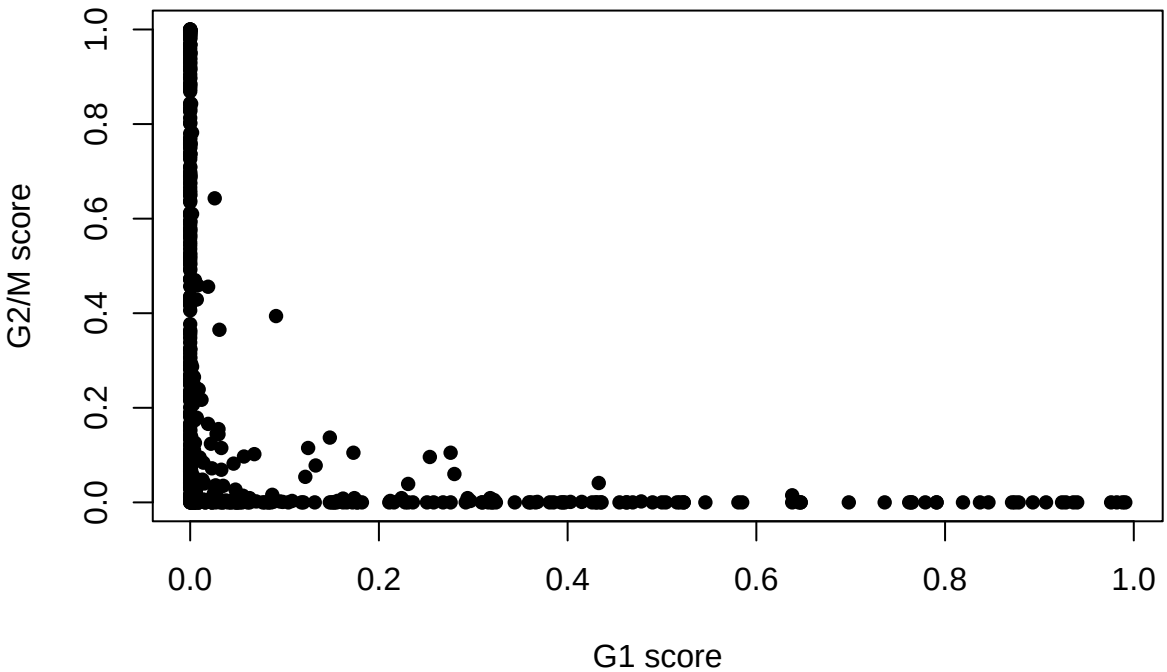
```
## 'select()' returned 1:many mapping between keys and columns
```

Cell Cycle Annotation

Cells in different phases of the cell cycle can show different expression profiles and different cell types may be on average in different phases of the cell cycle. The impact of this on our analysis if determined by assigning each cell to a cell cycle phase using the package Cyclone and cell cycle genes from the package Scran. Cells arw assigned to a unique phase of the cell cycle if they are predicted to assign to one phase of the cell cycle with a score >0.5. Cells that do not pass this criteria for one cell cycle stage only or not for any cell cycle phase are not assigned. The cell cycle information is added to a slot called CellCycle in the Single Cell Experiment (sce) object.

Note ambiguous are annotated as *None*

The plot shows G2/M assignments vs G1 for this data- S is also assigned (not shown)



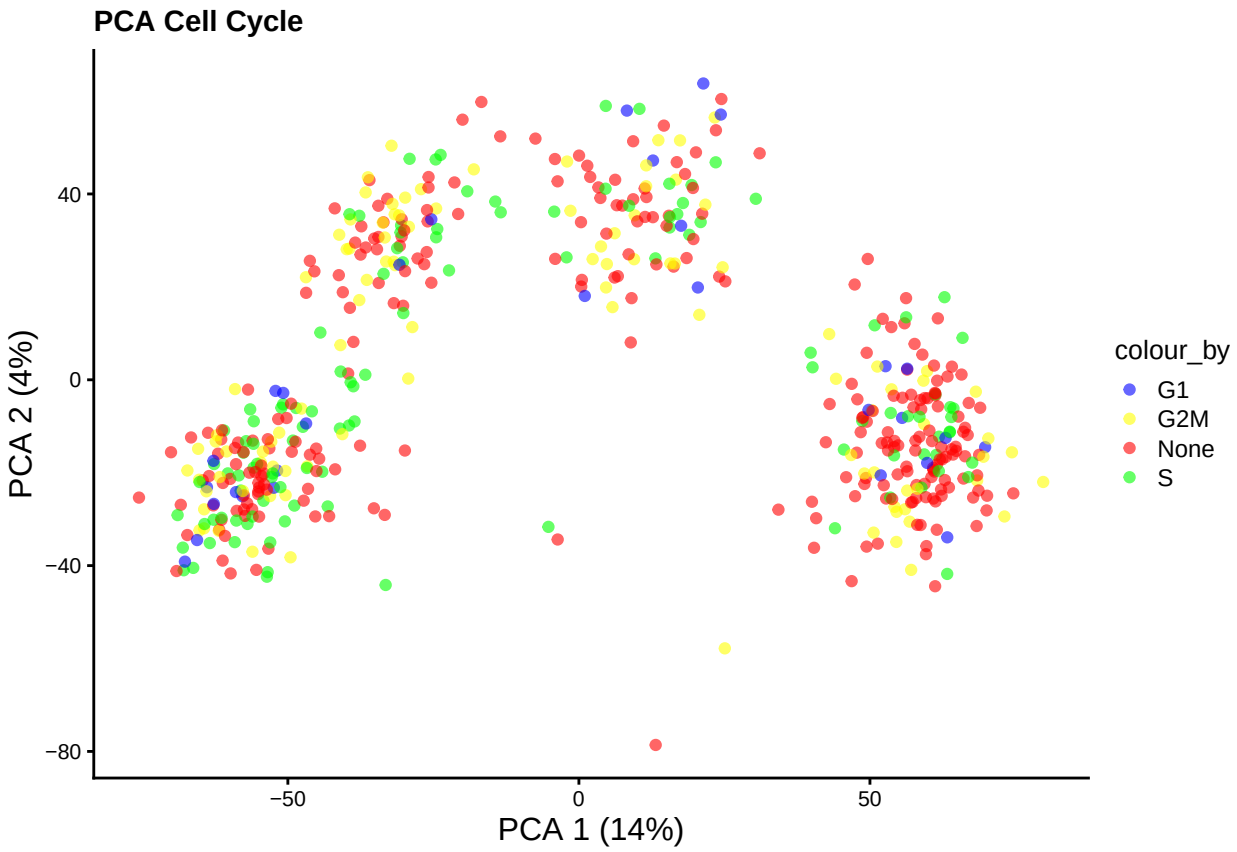

```
## [1] 515

## [1] 515

## [1] 515

## [1] 515

## [1] "character"

## [1] 515

## [1] "Added CellCycle slot to sce object"

##
##   G1  G2M None    S
##   30  102  268  115

## [1] 515
```

Evaluation of Cell Cycle Annotation

Output of Cell Cycle Predictions plotted onto a PCA plot of the sce data set which has been filtered for poor quality cells but not to remove genes with low read counts.

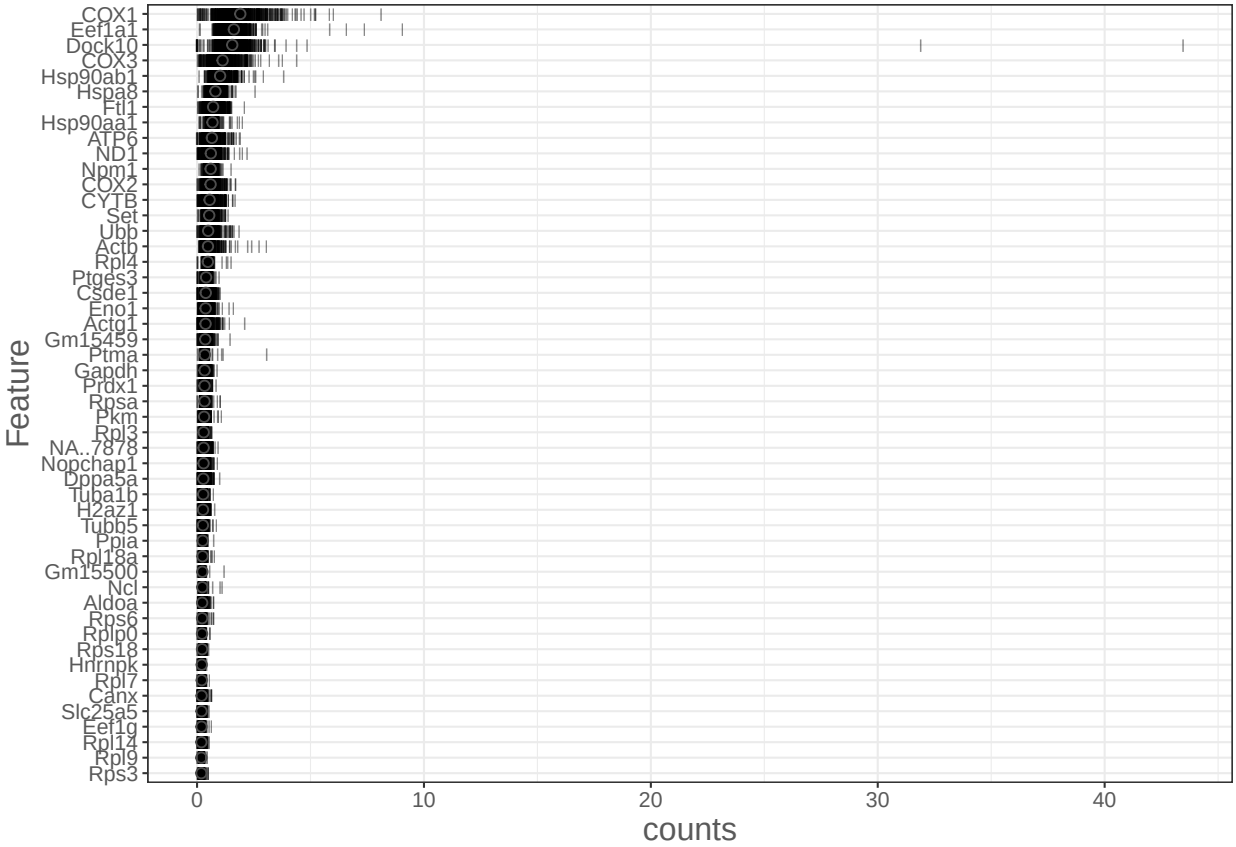
There is no obvious partitioning of the cell cycle phased between individual groups- all groups contain a fairly even mixture of phases.

```
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
```



Plot Top 50 Most Highly Expressed Genes from the Combined Data Set

It is important to see that the top 50 genes do not account for most of the sequencing counts! Remember that we expect "housekeeping" such as ribosomal proteins and genes involved in basic cellular functions



Remove the Spikes and Put These Into an Alt Experiment Within the Single Cell Experiment Object

This moves the spikes out of the main object so that they do not intefere with the normalisation of the main data. Spikes are normalised separately…

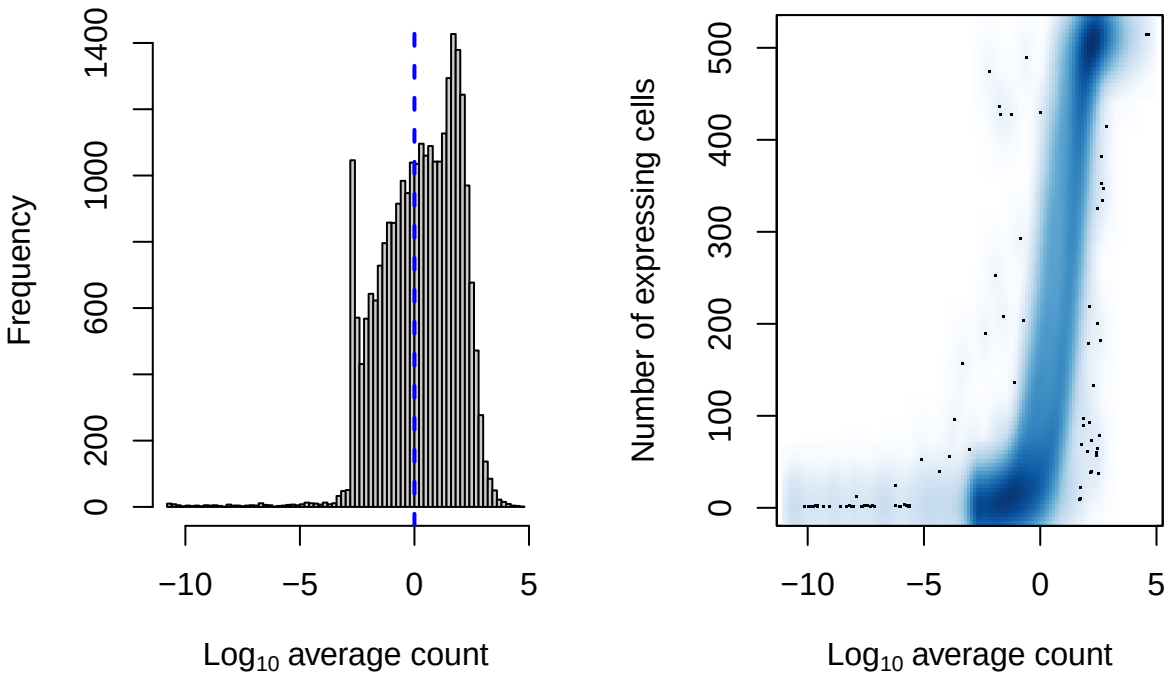
Remove Genes with Low Counts…

Several methods were considered but here filtering was simply by keeping cells with an average count across all genes of >=1

Filtering makes quite a difference to the analysis and actually several different gene filterings (including no gene filtering) were tried across the analysis before this simple approach was taken. In practice this will remove low expressed genes and genes expressed in few cells. These genes are not the target of this analysis..

```
## [1] 15528
```

```
## [1] 20730
```




```
## [1] "Number Genes Before: 32631"
```

```
## [1] "Number Genes After: 20730"
```

Remove Bulk Samples that May Still be Present in the Data

```
## [1] 5
```

```
## [1] 510
```

```
## [1] 510
```

```
## [1] "Bulk Samples Remaining"
```

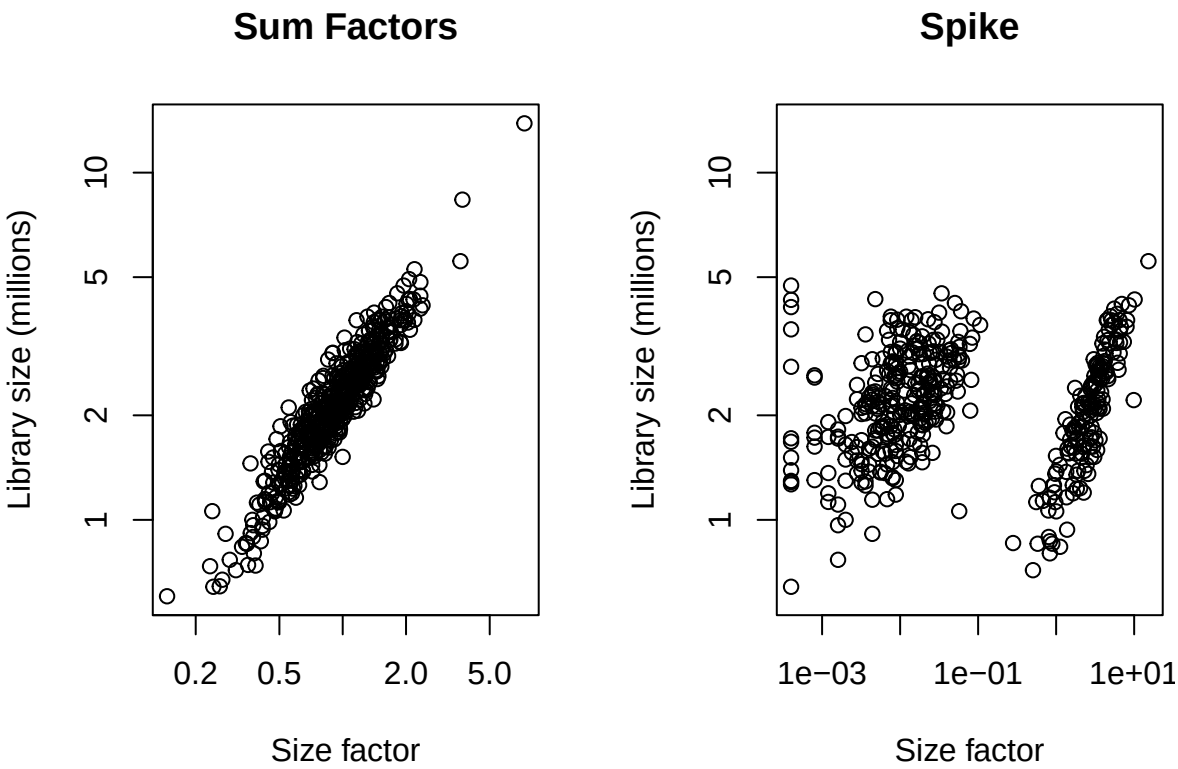
```
##
##    2i   a2i serum
##   243    89   178
```

Normalisation of Data

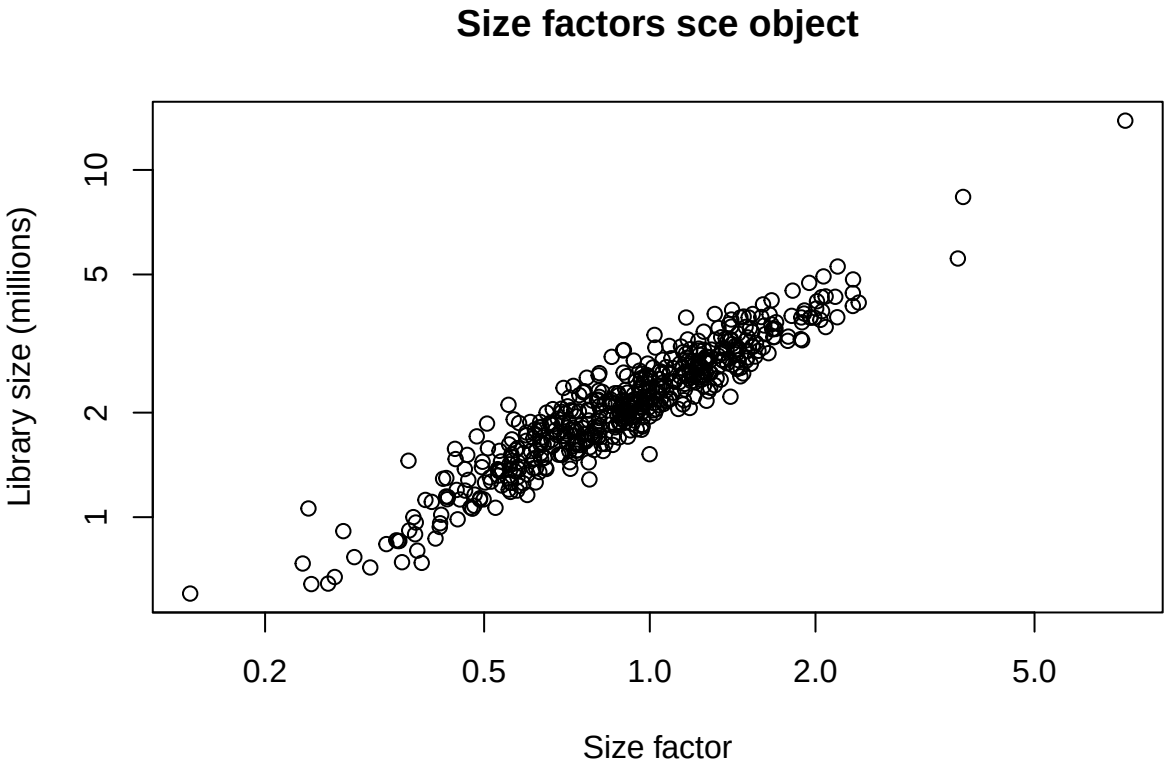
It is not clear if we should use spikes or the data distribution for normalisation. Data distribution uses the plate ID that contains each cell.

Normalisation would seek to identify sizeFactors for each cell library that are the scaling factors used to scale each library. ***

```
## Warning in xy.coords(x, y, xlabel, ylabel, log): 68 x values <= 0 omitted from
## logarithmic plot
```



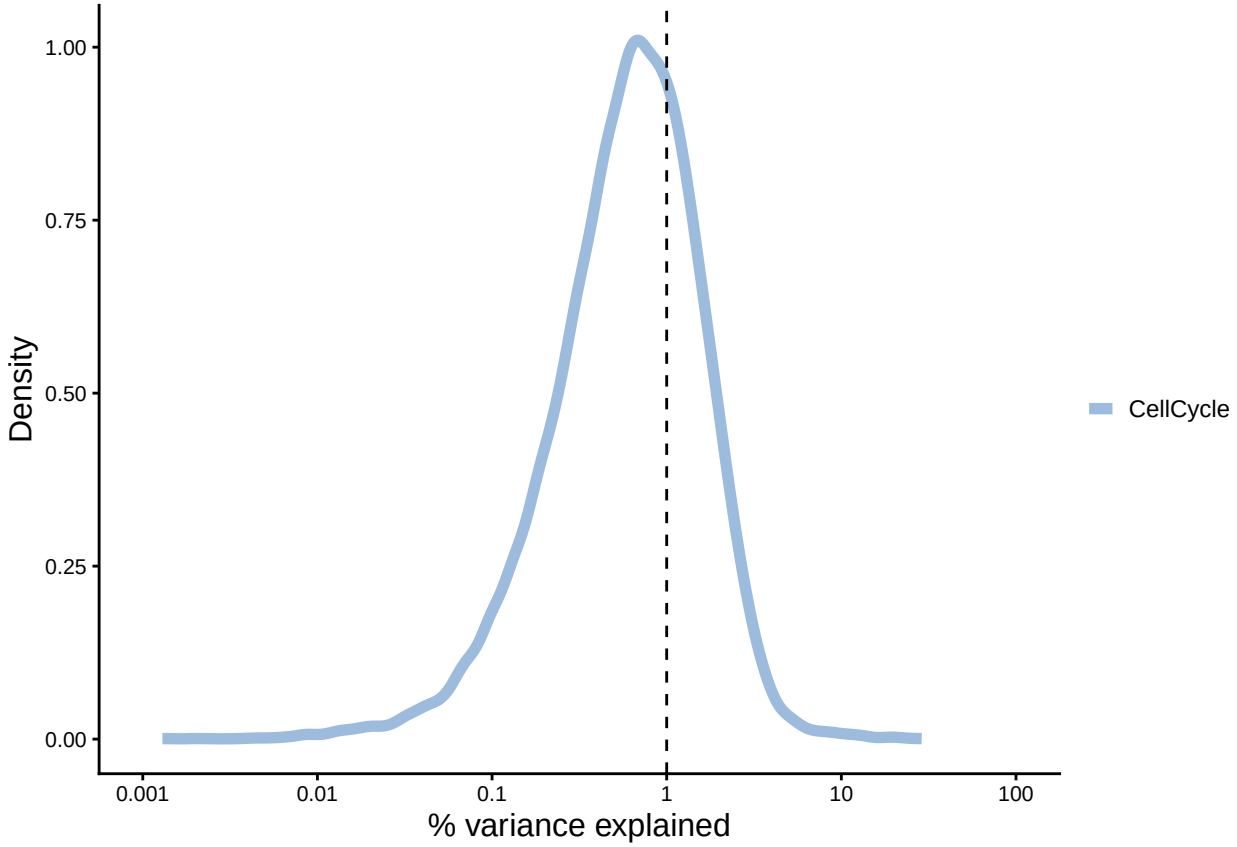

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.1462  0.6638  0.9107  1.0000  1.2311  7.3093
```



QC Feature counts for Explanatory Variables

This plot just shows some of the features by counts for explanatory variables-here Cell Cycle. Data is not log transformed

```
## [1] "stats"                "stats_selected_cells" "CellCycle"
## [4] "Type"                 "sizeFactor"
```



Identify the Highly Variable Genes from the Data.

Here a loess trendline is fitted to the data to show the mean variance trend

Elements are decomposed into whether the variance observed can be explained by the technical model or is not explained by this model and so if likely to come from the biology of individual genes. In this analysis spikes are plotted but not used for data analysis. Data from all the individual cell types are included in the analysis.

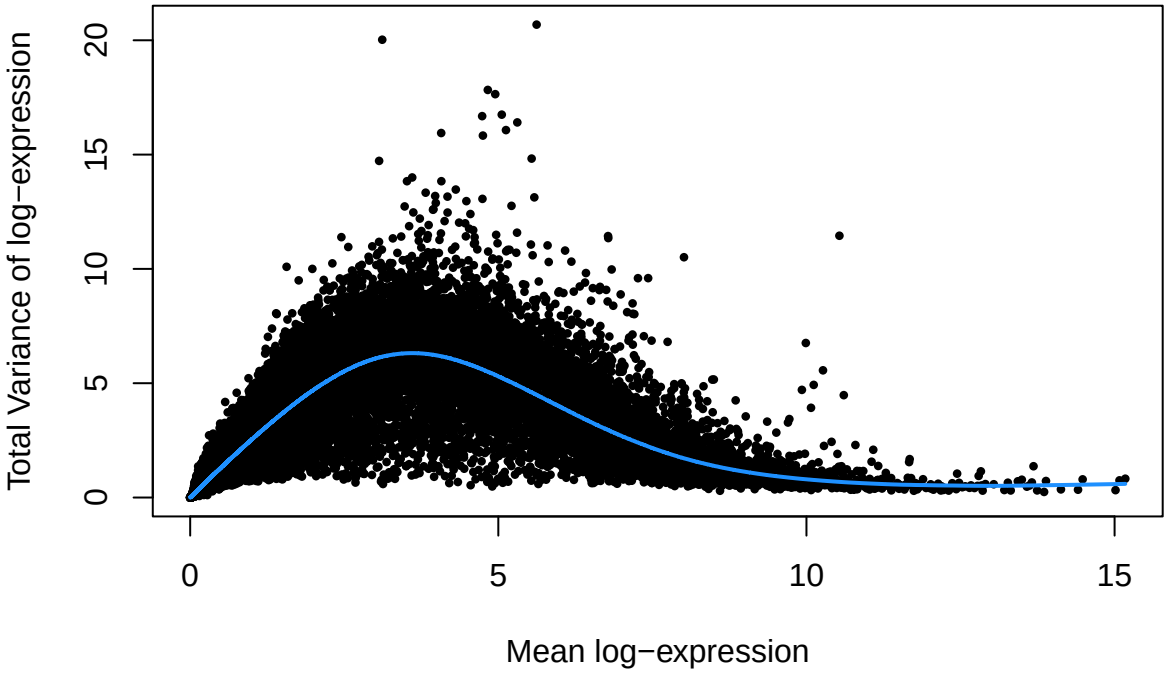
Initially HVG genes identified that are FDR <=0.2 and bio >0.5 as potential HGV genes (a very inclusive set) Output genes are ordered by decreasing biological variance. A tab-seperated table is written as "hsc_hvg.tsv".

The top 30 HVG genes (ordered by their biological expression component) are plotted as a "slug-plot" and typically show robust expression

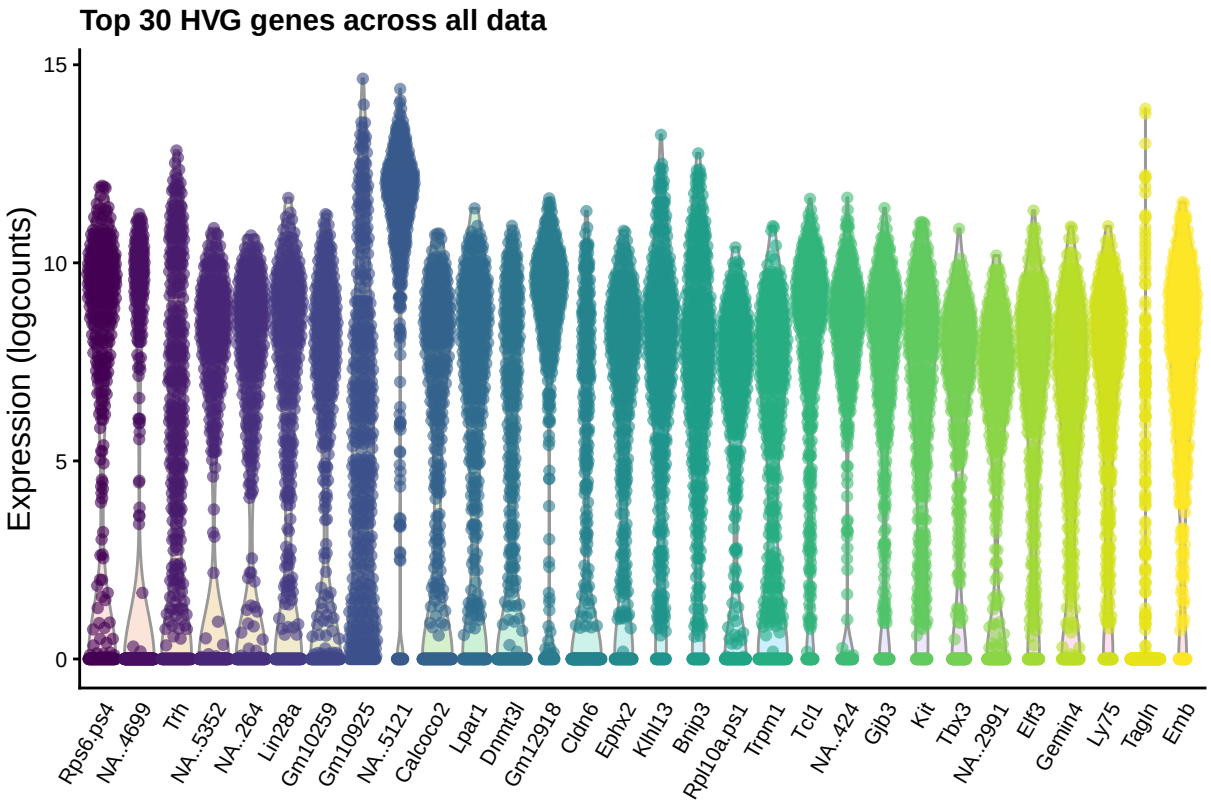
This data is also used for corrlation analysis and the correlated pairs of HGV genes identified. This table was output as "hsc_cor.tsv". Genes from this table with var.cor$FDR <= 0.01 and summarised. These genes become the "chosen" feature set used for future tSNE and PCA analysis

---

```
## Warning in regularize.values(x, y, ties, missing(ties), na.rm = na.rm):
## collapsing to unique 'x' values
```




```
## [1] 190

## DataFrame with 6 rows and 6 columns
##                mean     total      tech       bio     p.value         FDR
##           <numeric> <numeric> <numeric> <numeric>   <numeric>   <numeric>
## Rps6.ps4    5.62129   20.6842   4.49079   16.1934 6.56587e-16 1.51234e-12
## NA..4699    3.11648   20.0241   6.15922   13.8649 3.01797e-07 1.60417e-04
## Trh         4.82973   17.8276   5.50354   12.3240 3.45395e-07 1.74635e-04
## NA..5352    4.95023   17.6425   5.36213   12.2804 1.91932e-07 1.20568e-04
## NA..264     5.05606   16.7471   5.23296   11.5142 5.37112e-07 2.47430e-04
## Lin28a      5.30836   16.4073   4.91051   11.4968 1.05171e-07 6.81308e-05
```




```
## Warning in sqrt((1 - r^2)/(n - 2)): NaNs produced
```

```
## DataFrame with 6 rows and 5 columns
##          gene1       gene2       rho      p.value          FDR
##    <character> <character> <numeric>    <numeric>    <numeric>
## 1       Gm6763      Gm8764  1.000000  0.00000e+00  0.00000e+00
## 2      Gm11237       Gm428  0.887343 6.50406e-173 5.83837e-169
## 3        Kpna2     Gm10184  0.832060 4.02069e-132 2.40611e-128
## 4      Gm11756     Gm11758  0.792821 2.50698e-111 1.12519e-107
## 5   Obox4.ps28  Obox4.ps27  0.731179  2.02567e-86  7.27336e-83
## 6        Trpm1        Gpx1  0.717381  8.98401e-82  2.68816e-78
```

```
##    Mode   FALSE    TRUE    NA's
## logical   10394    7559       2
```

```
## [1] "number of chosen genes for plots"
```

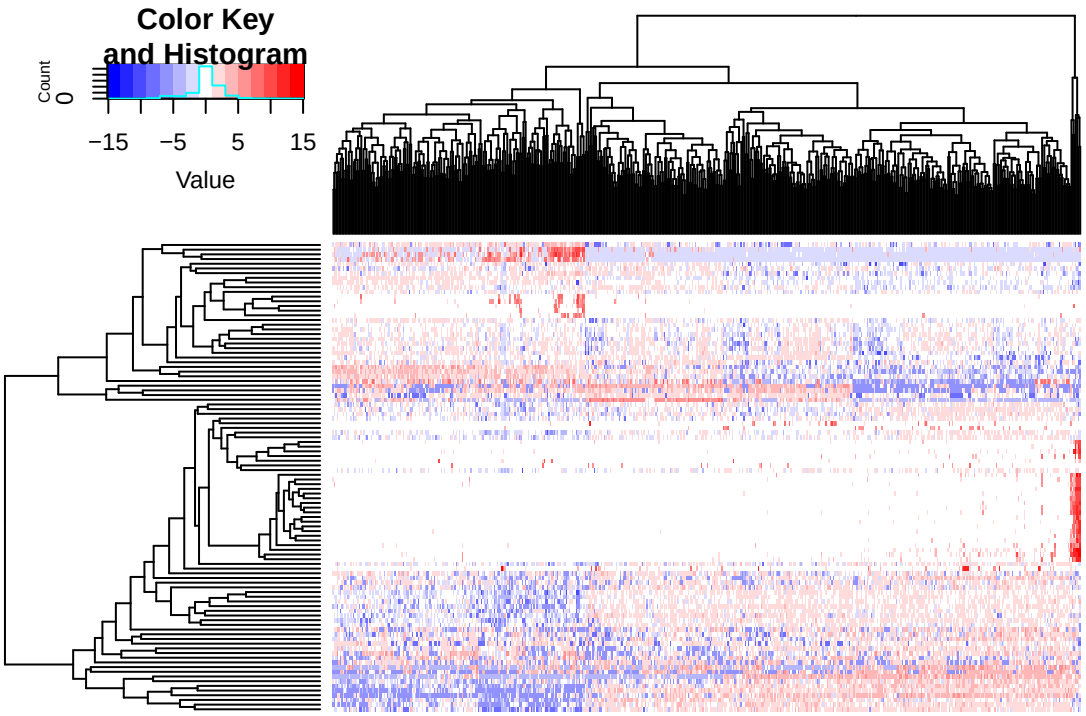
```
## [1] 190
```

```
## [1] "number of HVG genes for plots"
```

```
## [1] 6
```

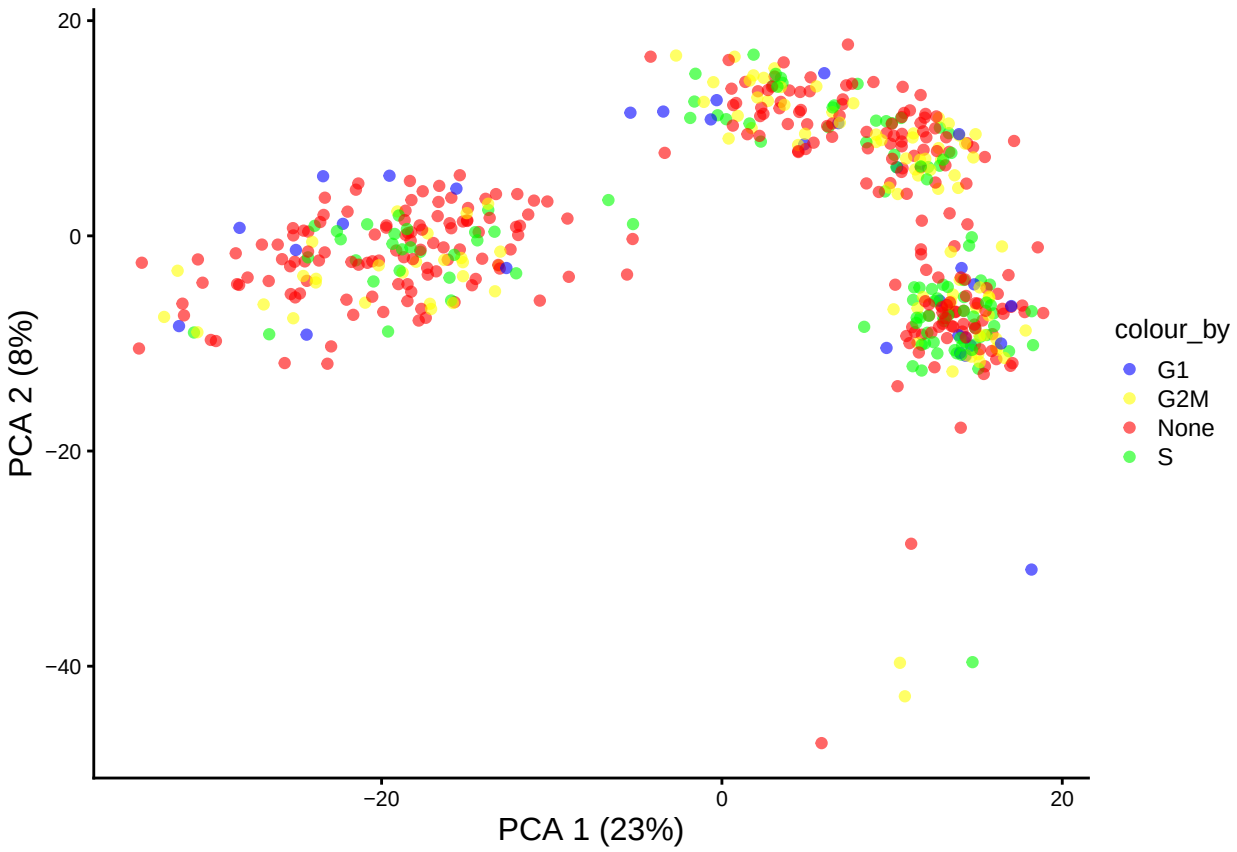
Make a Heatmap to Show a Random Sample of 100 of the Top HVG Genes

Genes are row mean centered



PCA of All Data From the Most Variable Genes Colouring by log total features using the chosen feature set. Note that this only accounts for ~14% of the variance in the plots. There are some clusters with higher levels of expression than others although note that the QC metric used is calculated before normalisation-normalisation should minimise the impact of this issue on the analysis.

```
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
```



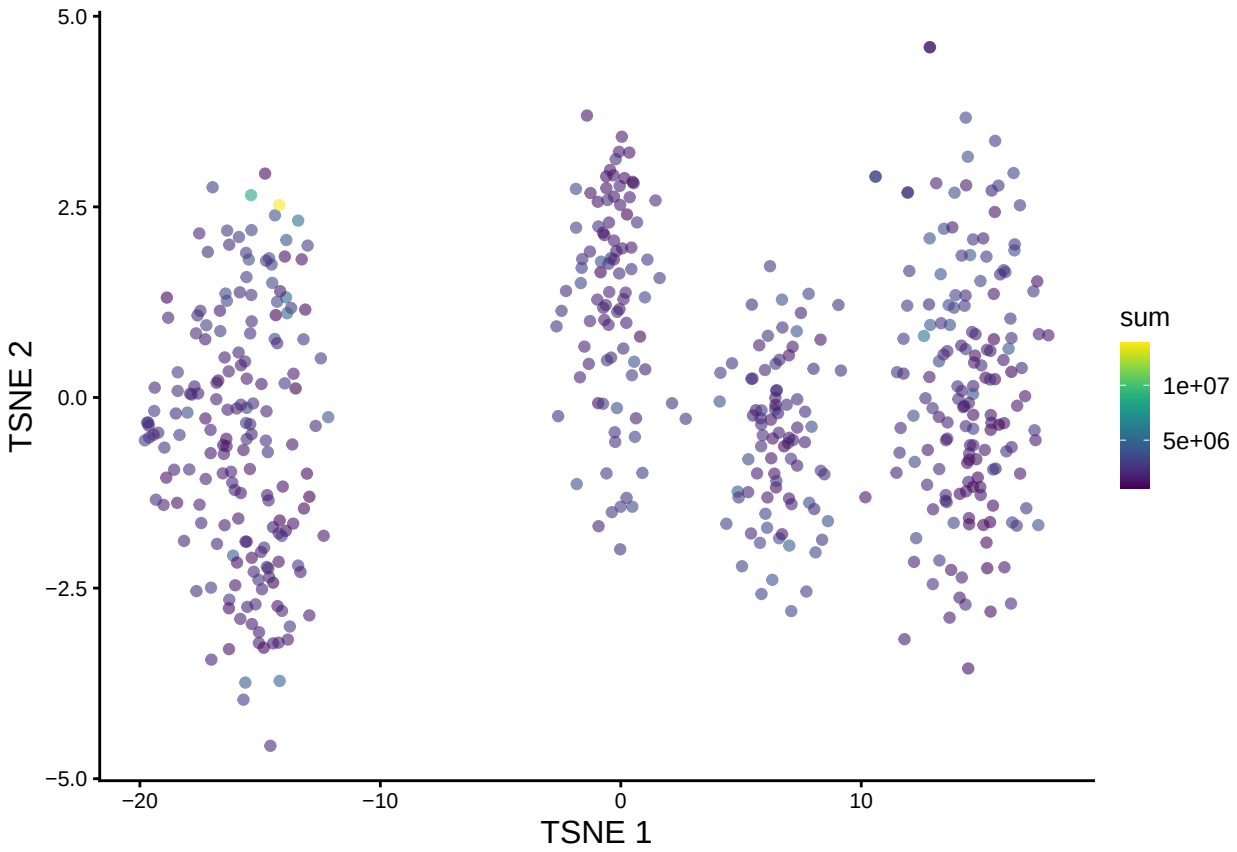
---

tSNE Plot Coloured by Total Features per Sample

tSNE is a dimensional reduction methods similar to PCA.

tSNE of all data using the set of most variable genes and colouring by log total features using the chosen feature set. Again this annotation is generated prior to normalisation

tSNE generates three large clusters but also several smaller clusters not visible in the PCA plot- tSNE will become the primary dimensional reduction method used in this analysis.



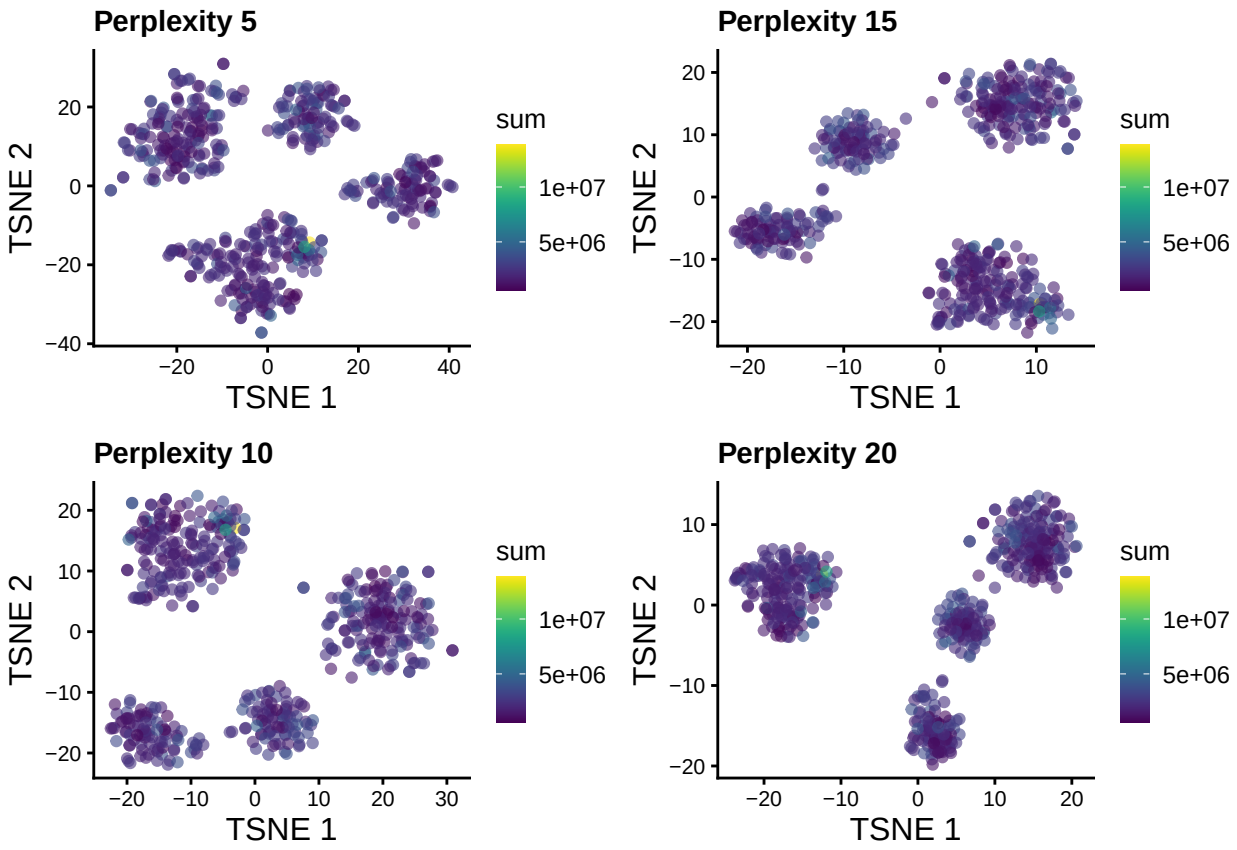
Selection of the tSNE 'Perplexity' Tuning Parameter

Dimensional reduction plots can give various results depending upon setting of tuning parameters. Here we are looking for that tSNE visualisation to be relatively stable within the recommended Perplexity range

We run the same scan again with a tSNE perplexity scan on total count data…5,10,15,20 Note these plots are not saved to the sce object

Plots are relatively stable. This suggests that the features detected in the data are relatively robust. We chose to run with perplexity =10 for this analysis

```
## Warning: 'multiplot' is deprecated.
## Use 'gridExtra::grid.arrange' instead.
## See help("Deprecated")
```



Loading 'Target' File and Building Sample Annotation

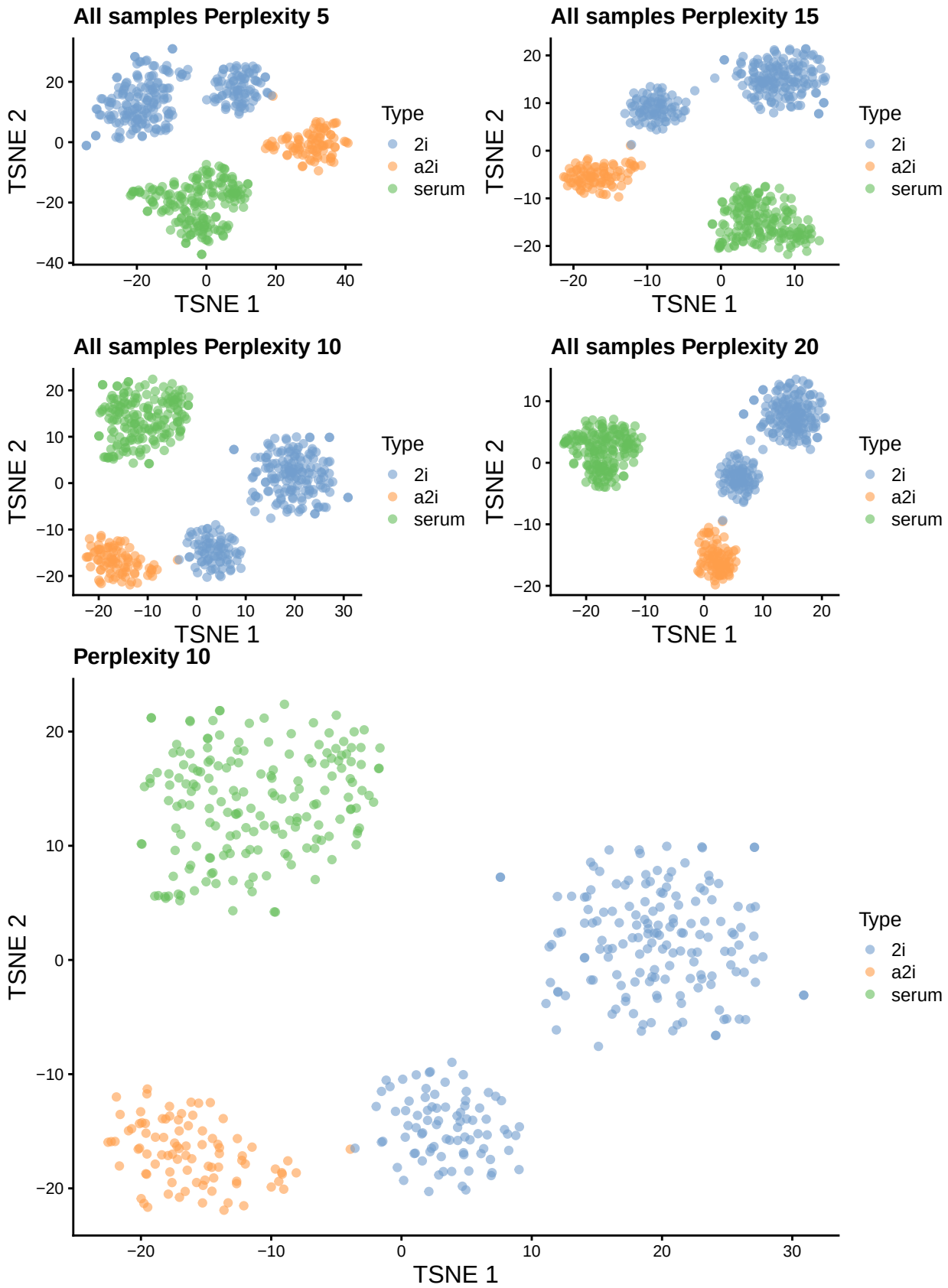
Load targets file and build this annotation into the SingleCellExperiment object.

```
## [1] "Column names of loaded \"targets\" file"
```

```
## [1] "Source.Name"  "ERS"          "Type"         "MaterialType" "ExtractName"
## [6] "AssayName"    "ERX"          "ERR"
```

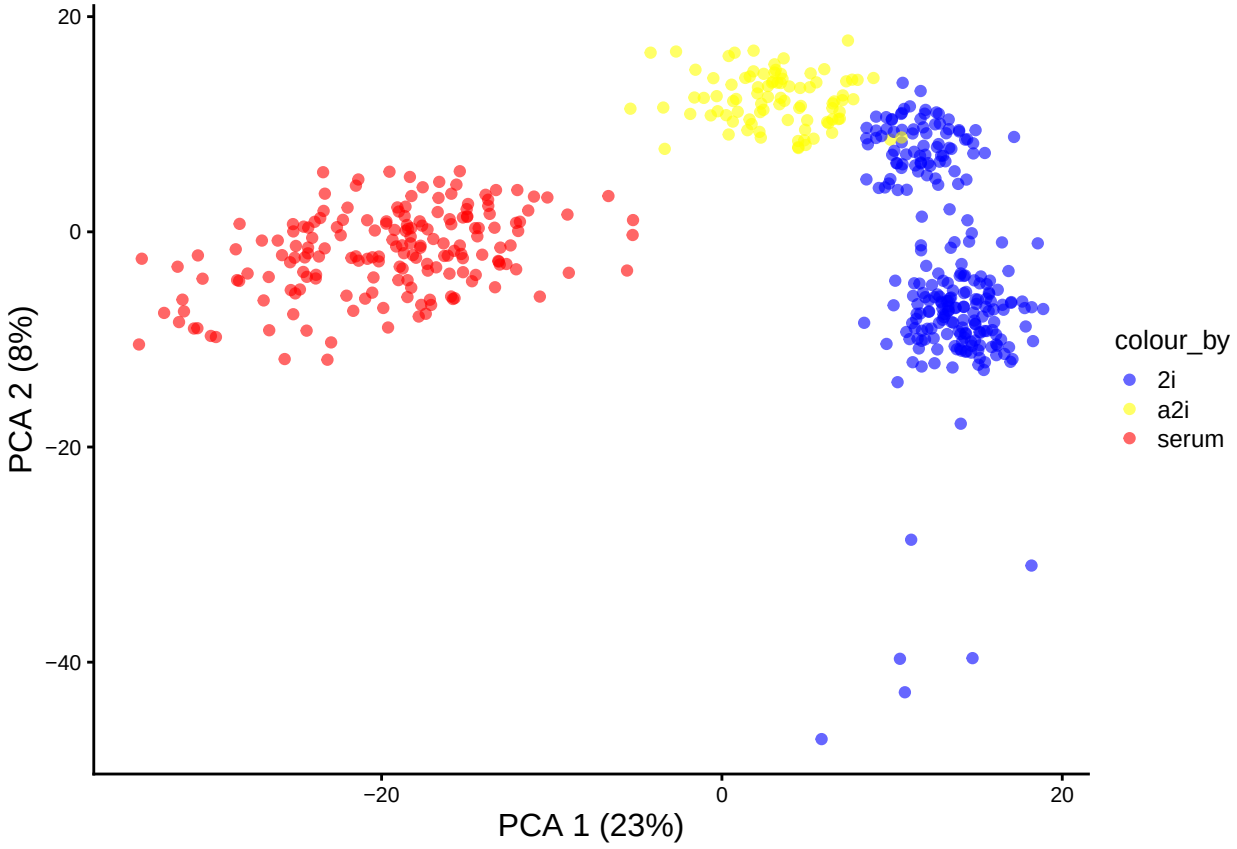
```
## [1] 0
```

```
## Warning: 'multiplot' is deprecated.
## Use 'gridExtra::grid.arrange' instead.
## See help("Deprecated")
```


All samples Perplexity 5
All samples Perplexity 15
All samples Perplexity 10
All samples Perplexity 20
Perplexity 10
TSNE 1
TSNE 2
Type
2i
a2i
serum

Plot final PCA of All Data Coloured by Type Loaded From the Targets File

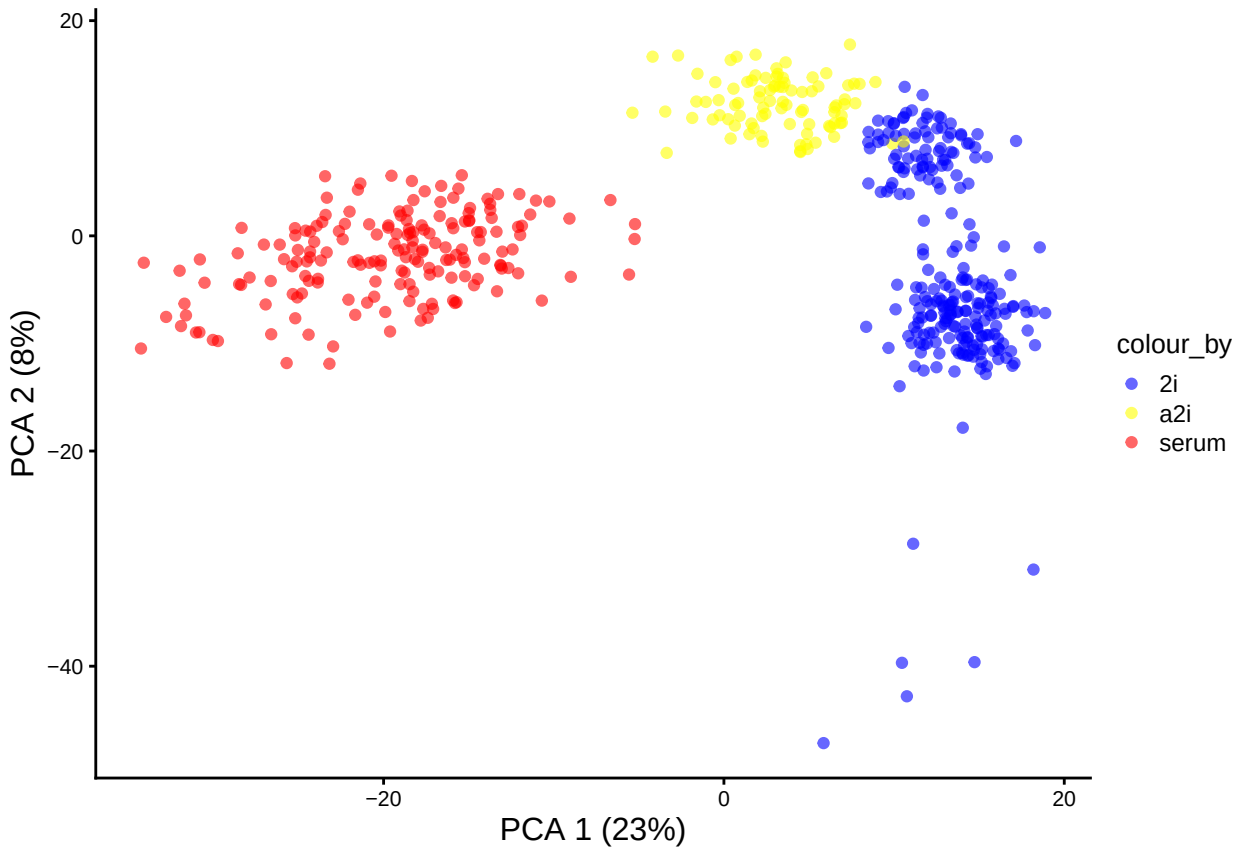
```
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
```



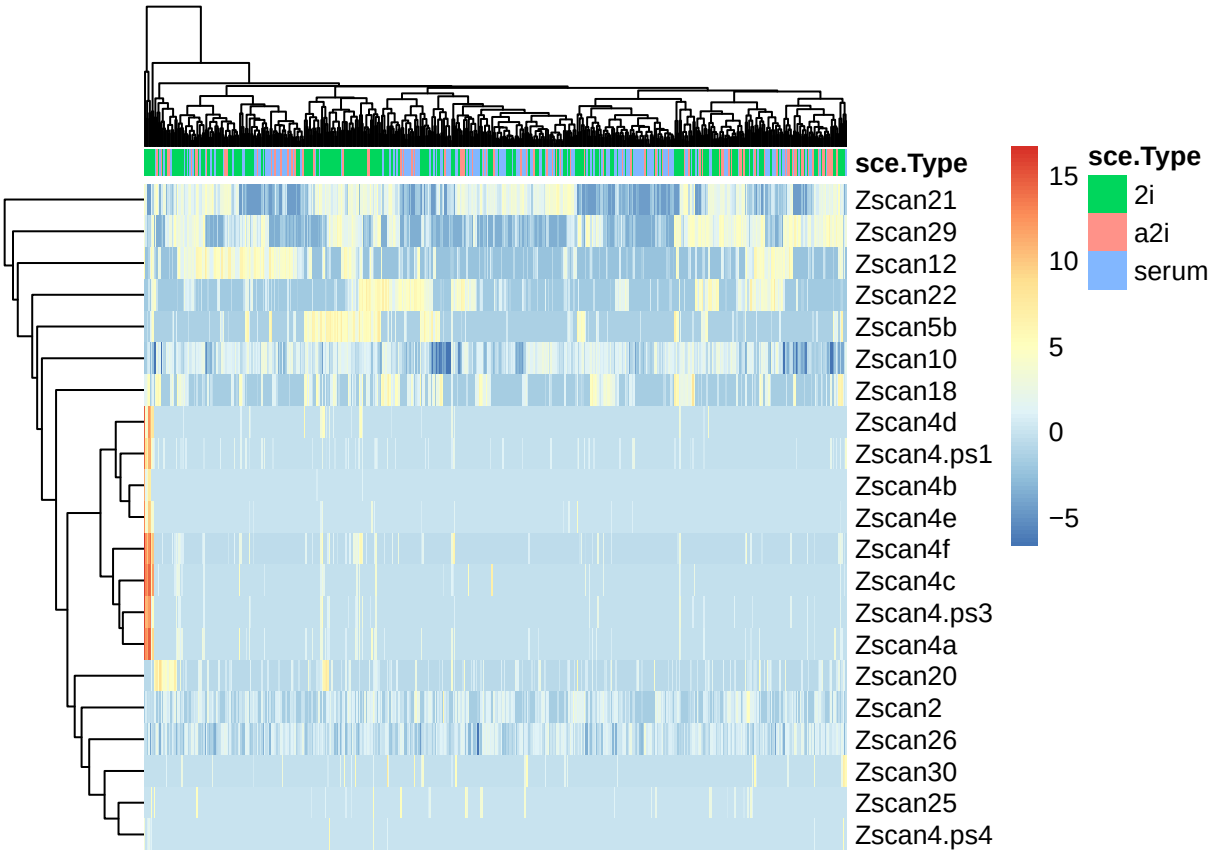
Final PCA Plot With All SCE Data

Again this uses all the data

```
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
```



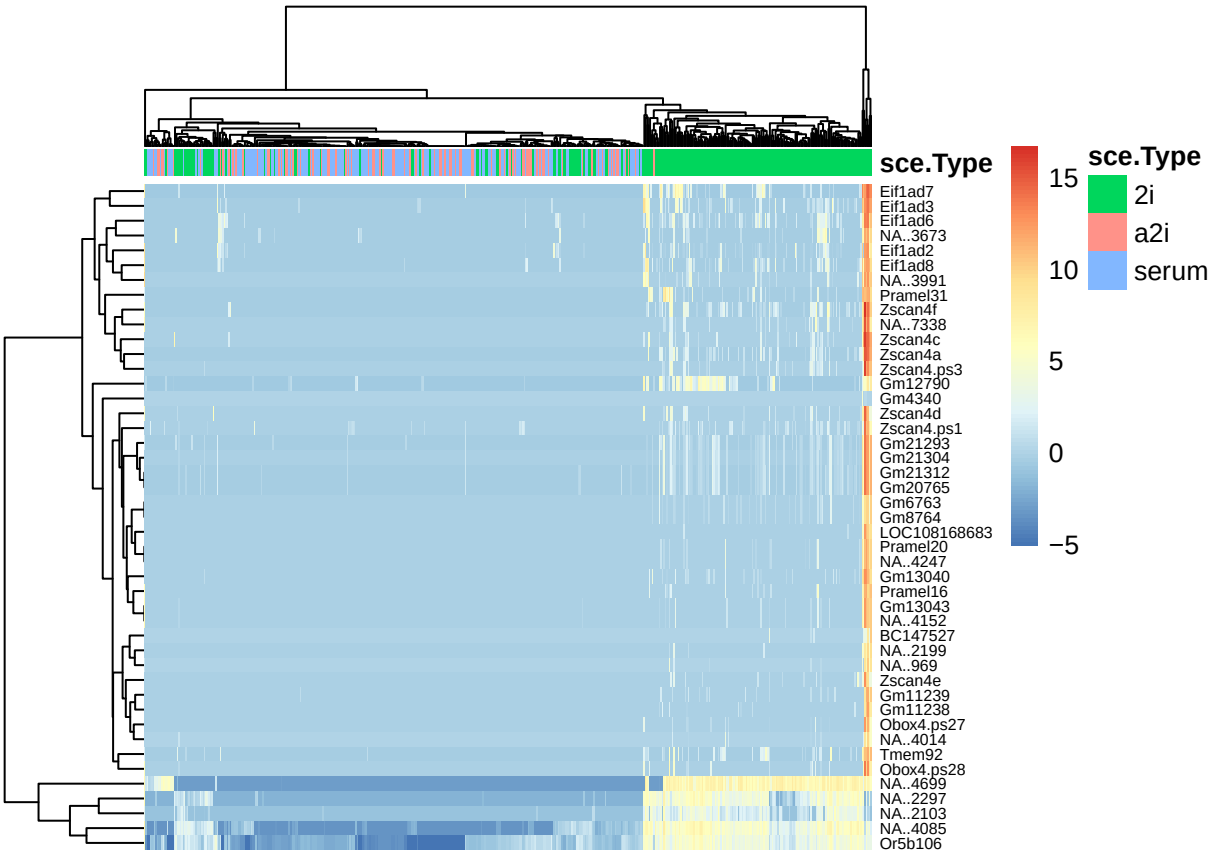
#Examine all ZScan Genes in the Expression Data ..



All ZScan Correlated Genes in the Data

Examine all genes in the expression data by using correlation to one ZScan profile. This finds genes that have a similar profile to Zscan4a but are not necessarily annotated as ZScan genes.

```
## keep_me
## FALSE  TRUE
## 20685    45
```


sce.Type
Eif1ad7
Eif1ad3
Eif1ad6
NA..3673
Eif1ad2
Eif1ad8
NA..3991
Pramel31
Zscan4f
NA..7338
Zscan4c
Zscan4a
Zscan4.ps3
Gm12790
Gm4340
Zscan4d
Zscan4.ps1
Gm21293
Gm21304
Gm21312
Gm20765
Gm6763
Gm8764
LOC108168683
Pramel20
NA..4247
Gm13040
Pramel16
Gm13043
NA..4152
BC147527
NA..2199
NA..969
Zscan4e
Gm11239
Gm11238
Obox4.ps27
NA..4014
Tmem92
Obox4.ps28
NA..4699
NA..2297
NA..2103
NA..4085
Or5b106
15
10
5
0
−5
sce.Type
2i
a2i
serum